A Project Report

on

Personalized News Aggregator with LLM Based Summarization and Fake News Detection

Submitted to the

Savitribai Phule Pune University

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Bachelor of Engineering

in

Information Technology

By

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Abstract

In today's digital age, the exponential growth of information has created an overwhelming landscape of news content, making it increasingly difficult for users to access relevant, accurate, and trustworthy information. The proliferation of fake news and misinformation has further complicated the news consumption process, creating an urgent need for intelligent systems that can filter, verify, and personalize news content effectively.

This project presents the development of a "Personalized News Aggregator with LLM Based Summarization and Fake News Detection" - an intelligent system designed to address the challenges of information overload and misinformation in digital news consumption. The system leverages cutting-edge technologies including Large Language Models (LLMs), Natural Language Processing (NLP), and Machine Learning algorithms to create a comprehensive solution for modern news consumption.

The primary objective of this project is to develop an automated news aggregation platform that collects news articles from multiple reliable sources, processes them through advanced NLP techniques, and delivers personalized, summarized, and verified content to users. The system employs LLMs for generating concise, coherent summaries of lengthy news articles, enabling users to quickly grasp essential information without reading entire articles.

1. Introduction

In the modern digital era, the consumption of news and information has undergone a dramatic transformation. The internet has democratized access to information, enabling anyone with a connection to access news from countless sources around the globe. However, this unprecedented access has created new challenges that threaten the quality and reliability of information consumption. Users today face an overwhelming deluge of news articles, blog posts, social media updates, and multimedia content, making it increasingly difficult to identify credible sources, extract relevant information, and stay informed without investing significant time and cognitive effort.

The proliferation of misinformation and fake news has emerged as one of the most pressing challenges of our time. False narratives spread rapidly across digital platforms, influencing public opinion, affecting democratic processes, and eroding trust in legitimate journalism. Traditional fact-checking methods, while valuable, cannot scale to match the exponential growth of online content. Furthermore, the personalization algorithms employed by major platforms often create echo chambers, reinforcing existing beliefs while limiting exposure to diverse perspectives.

1.1 Aim

The primary aim of this project is to design, develop, and implement an intelligent, automated news aggregation system that leverages Large Language Models (LLMs) and Machine Learning techniques to provide users with personalized, summarized, and authenticity-verified news content, thereby addressing the critical challenges of information overload, misinformation proliferation, and inefficient news consumption in the digital age.

This aim encompasses:

- Combating Misinformation: Developing robust fake news detection mechanisms to filter out unreliable and false information before it reaches users.
- Addressing Information Overload: Implementing intelligent summarization techniques that condense lengthy articles into concise, coherent summaries while preserving essential information.
- Enhancing Personalization: Creating adaptive recommendation systems that understand individual user preferences and deliver relevant content without creating echo chambers.
- Improving Accessibility: Making quality journalism and reliable information accessible to users with varying time constraints and reading preferences.

1.2 Objective

1. Develop automated web scraping modules for multi-source data collection.
2. Build machine learning models for fake news detection with ≥85% accuracy.
3. Integrate transformer-based LLMs for 70–80% content reduction while retaining 90% information.
4. Create a user profile and recommendation engine for personalized news feeds.
5. Implement scalable microservices architecture with a responsive web interface.

2. Literature Survey

Sr. No.	Paper Details	Problem Discussion	Algorithm /Technique used	Parameter Consider	Result
1	"BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding" by Devlin et al., 2018	Traditional NLP models fail to capture bidirectional context in text understanding	Bidirectional Encoder Representations from Transformers (BERT)	Masked Language Modeling, Next Sentence Prediction, ng Batch Size	Achieved state-of-the-art results on 11 NLP tasks with significant improvements
2	"FakeNews-Net: Repository with News, Context and Spatial- for Studying Fake on Media" by Shu et al., 2018	Lack of comprehensive datasets for fake news detection research	Graph-based analysis, Content-based Social context analysis	Content features, Social engagements, User profiles, Temporal patterns	Created comprehensive benchmark dataset with 23,196 news articles
3	"Text Summarization Techniques: A Brief Survey" by Allahyari et al., 2017	Existing summarization methods lack coherence and fail to preserve key information	Extractive and Abstractive , TextRank, Sequence-to-Sequence models	ROUGE scores, Coherence metrics, Computational complexity	Comparative analysis showing abstractive methods achieve better coherence
4	"A Survey on Neural News Recommendation" by Wu et al., 2021	Traditional recommendation systems fail to capture complex user preferences in news domain	Deep learning models, mechanisms, Multi-task learning	User embeddings, News ddings, Click-through rates, Dwell time	Neural methods outperform traditional collaborative filtering by 15-20%
5	"Combating Fake News: A on and Techniques" by Zhou et al., 2019	Manual fact-checking cannot scale with the volume of online content	Machine Lerning, approaches, Ensemble methods	Linguistic features, Source Propagation patterns, Network analysis	Ensemble methods achieve 90%+ accuracy in controlled datasets

3. Problem Statement/definition

In the contemporary digital information ecosystem, users face unprecedented challenges in accessing reliable, relevant, and concise news content. The exponential growth of online news sources, coupled with the proliferation of misinformation and the absence of effective personalization mechanisms, has created a critical need for intelligent systems that can filter, verify, summarize, and personalize news content effectively.

The core problem this project addresses is: "How can we develop an automated, intelligent news aggregation system that simultaneously combats misinformation, addresses information overload through intelligent summarization, and delivers personalized content tailored to individual user preferences while maintaining credibility, diversity, and accessibility?"

This problem encompasses multiple interconnected challenges that collectively impede effective news consumption in the digital age, requiring a comprehensive, integrated solution leveraging advanced artificial intelligence and natural language processing technologies.

The digital age has witnessed an unprecedented explosion in content creation. According to recent estimates, over 2.5 million blog posts are published daily, and major news organizations collectively produce thousands of articles across multiple platforms every hour. This abundance of information, while democratizing access to knowledge, has created a paradoxical situation where users have access to more information than ever before but struggle to extract meaningful insights from this deluge.

The spread of misinformation, disinformation, and fake news has emerged as one of the most significant threats to informed decision-making and democratic discourse. False narratives spread faster and further than accurate information, exploiting cognitive biases and emotional triggers to achieve viral propagation across social networks.

Generic news feeds fail to account for the diverse interests, expertise levels, and information needs of individual users. The one-size-fits-all approach to news delivery results in poor user engagement, reduced satisfaction, and inefficient information consumption.

4. Software Requirement Specification

A] System Overview

The Personalized News Aggregator is an intelligent web-based platform that leverages Large Language Models and Machine Learning to deliver verified, summarized, and personalized news content. The system automatically collects news from multiple reliable sources, analyzes content authenticity, generates concise summaries, and recommends articles based on individual user preferences.

Main Modules:

1. **News Aggregation Module** - Automated collection from multiple sources
2. **Fake News Detection Module** - ML-based content verification
3. **LLM Summarization Module** - Intelligent text summarization
4. **Personalization Engine** - Adaptive recommendation system
5. **User Management Module** - Profile and preference handling
6. **Search & Discovery Module** - Advanced filtering and search
7. **Dashboard & Analytics** - User insights and reporting

B] Users

End Users / Readers:

- Register and create personalized profiles
- Access curated, summarized news feed

System Administrator:

- Manage news sources and categories
- Monitor system performance

Content Moderator (Optional):

- Review flagged articles
- Update source credibility scores

C] Functional Requirements

User Management:

- User registration with email verification
- Secure login/logout with JWT authentication

News Aggregation:

- Collect articles from 15+ reliable sources
- RSS feed integration for real-time updates

Fake News Detection:

- ML-based analysis using SVM and Random Forest algorithms
- Assign credibility scores (0-100) to articles and sources

Text Summarization:

- Generate abstractive summaries using BERT/GPT models
- Three summary lengths: Brief (2-3 sentences), Medium (1 paragraph), Detailed (2-3 paragraphs)

D] Non-Functional Requirements

Requirement	Description
Performance	Support 1,000+ concurrent users; Page load time <2 seconds; Process 1,000+ articles daily; Database queries <500ms
Security	AES-256 encrypted passwords; JWT token authentication; HTTPS communication; SQL injection & XSS protection; Secure API endpoints; GDPR compliance
Reliability	99.5% system uptime; Automated daily backups; Error logging and monitoring; Auto-recovery from failures within 5 minutes
Usability	Intuitive, responsive UI; Easy navigation; Mobile-friendly

Scalability	Microservices architecture; Horizontal scaling capability; Cloud deployment ready; Load balancing support; Handle growing user base
Maintainability	Modular code structure; Comprehensive documentation; Git version control; Automated testing; Easy deployment process
Accuracy	Fake news detection: 85%+ accuracy; Summary quality: 90% information retention; Recommendation relevance: 80%+ satisfaction

E] Technologies Used

Frontend:

- React.js 18
- HTML5, CSS3
- JavaScript ES6+
- Tailwind CSS

Backend:

- RESTful APIs
- JWT Authentication

Database:

- PostgreSQL 14 (structured user data)
- Redis 6.2 (caching, session management)

Machine Learning / AI:

- Transformers (Hugging Face) - BERT, GPT models
- TensorFlow / PyTorch
- Scikit-learn (SVM, Random Forest)
- NLTK, spaCy (NLP processing)
- Pandas, NumPy (data processing)

APIs & Integration:

- OpenAI API (LLM integration)
- News API (news aggregation)
- MediaStack API (news feeds)

Web Scraping:

- BeautifulSoup4
- Scrapy 2.6

Development Tools:

- Docker (containerization)
- Git (version control)
- Postman (API testing)
- Jupyter Notebook (ML experimentation)

Deployment:

- AWS / Azure (cloud hosting)
- Nginx (web server)

F] Future Enhancements

Multi-language Support:

- Extend beyond English to support Hindi, Marathi, and other Indian languages
- Automatic language detection and translation

Advanced Features:

- Video news summarization using computer vision
- Podcast and audio news integration

5. Flowchart

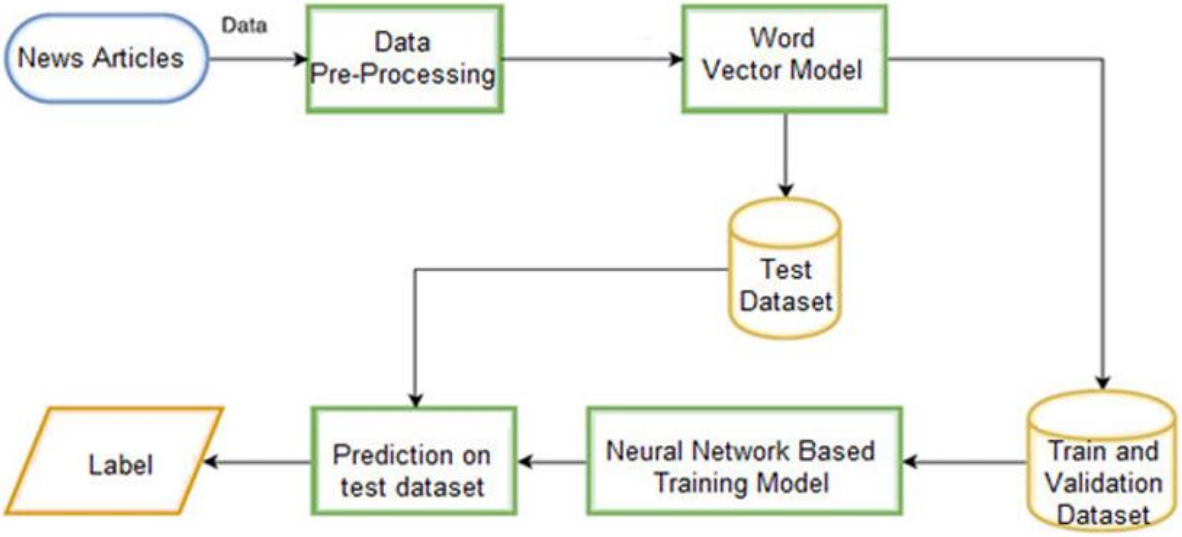

FIGURE 4. The General DL Framework that Used for FND.

Fig 5.1 Flowchart Diagram

6. Project Requirement specifications

Software Requirements:

Operating System:
- Windows 10/11 or Ubuntu 20.04 LTS or later

Programming Languages:
- Python 3.9+
- JavaScript ES6+

Frameworks/Libraries:
1. **Frontend:** React.js 18, HTML5, CSS3, Tailwind CSS
2. **Backend:** Flask 2.0 / FastAPI, Flask-JWT-Extended
3. **ML/AI:** Transformers (Hugging Face), TensorFlow/PyTorch, scikit-learn (SVM, Random Forest)
4. **NLP:** NLTK 3.7+, spaCy 3.4+, Pandas, NumPy
5. **Web Scraping:** BeautifulSoup4, Scrapy 2.6, Selenium

Database:
- MongoDB 5.0 (for articles)
- PostgreSQL 14 (for users)
- Redis 6.2 (for caching)

APIs:
- OpenAI API (LLM integration)
- News API, MediaStack API (news aggregation)

Other Tools:
1. Docker (containerization)
2. Git (version control)
3. Postman (API testing)
4. VS Code / PyCharm (IDE)
5. Jupyter Notebook (ML experimentation)

Hardware Requirements:

Development Environment:

Processor:
- Intel Core i7 8th gen or higher / AMD Ryzen 7 or higher

RAM:
- Minimum: 16 GB DDR4
- Recommended: 32 GB (for ML model training)

Storage:
- Minimum: 256 GB SSD
- Recommended: 1 TB SSD

Graphics (GPU):
- NVIDIA RTX 3060 with 8 GB VRAM or better (recommended for ML training)
- NVIDIA GTX 1660 Ti with 6 GB VRAM (minimum)

Network:

- High-speed internet (minimum 100 Mbps) for API integrations and updates

Production/Deployment Server:

Specifications:

- CPU: 4-8 cores
- RAM: 16-32 GB
- Storage: 500 GB - 1 TB SSD
- Bandwidth: Unlimited / High bandwidth
- Cloud: AWS / GCP / Azure

7. Proposed system Architecture

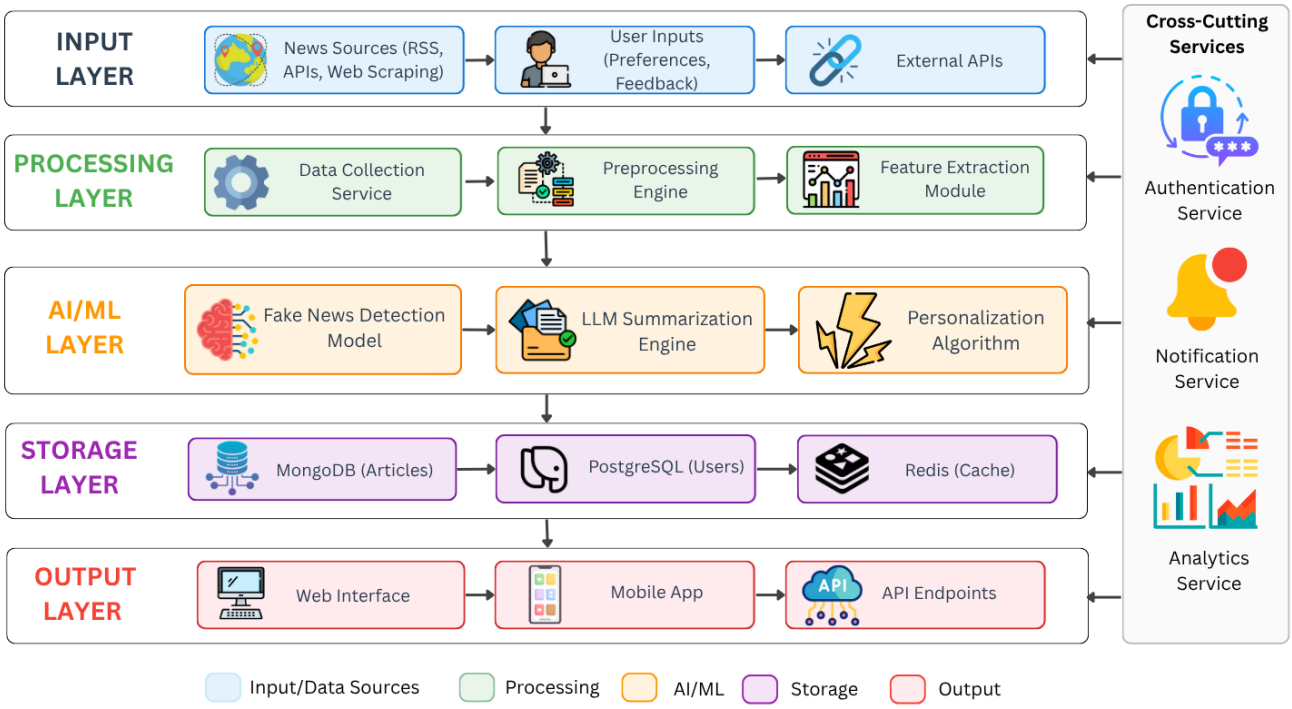

Fig 7.1 System Architecture Diagram

8. High level design of the project

- **High level Architecture**

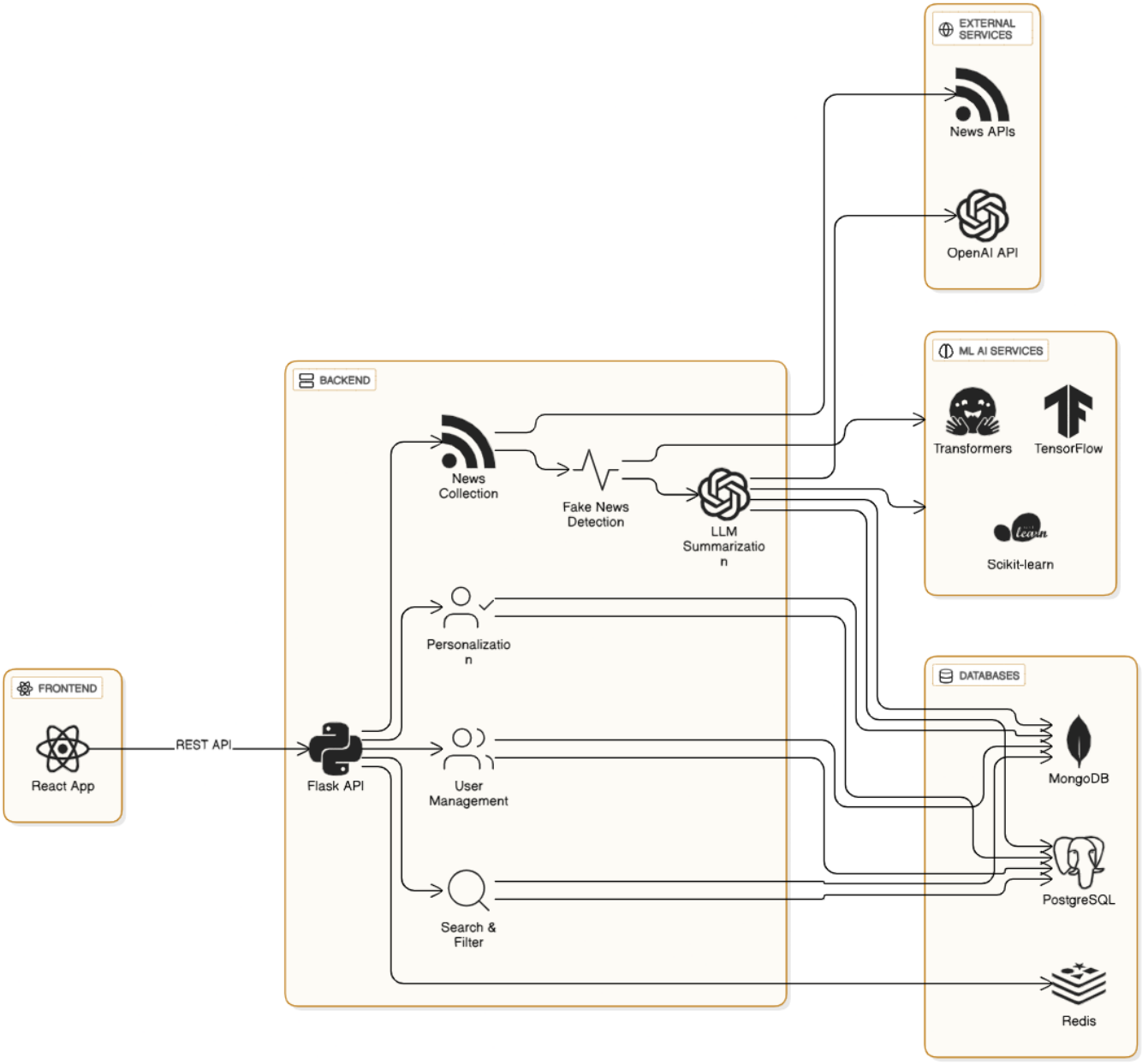

Fig 8.1 High Level Architecture Diagram

- **ER Diagram**

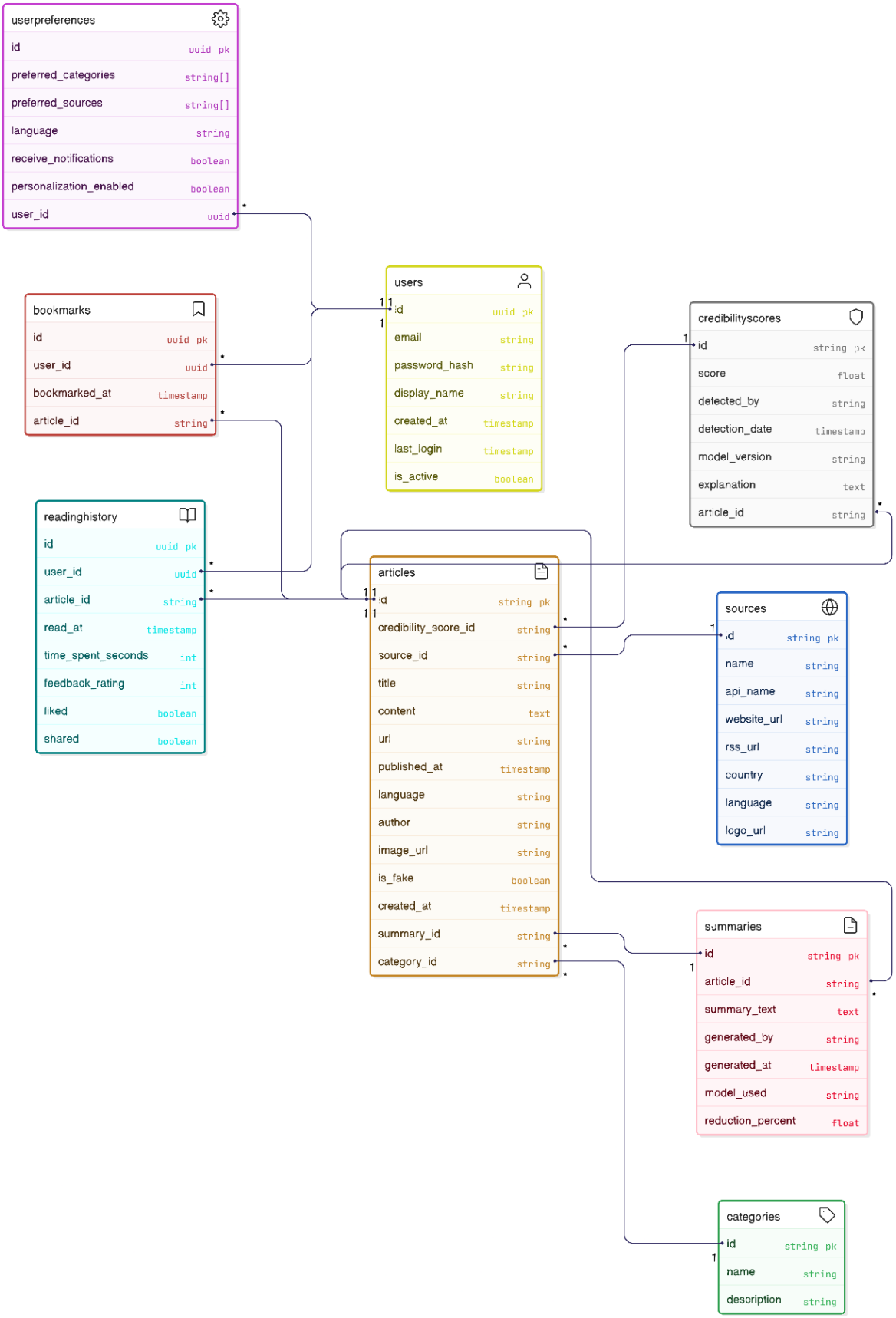

Fig 8.2 ER Diagram

- **Use Case Diagram**

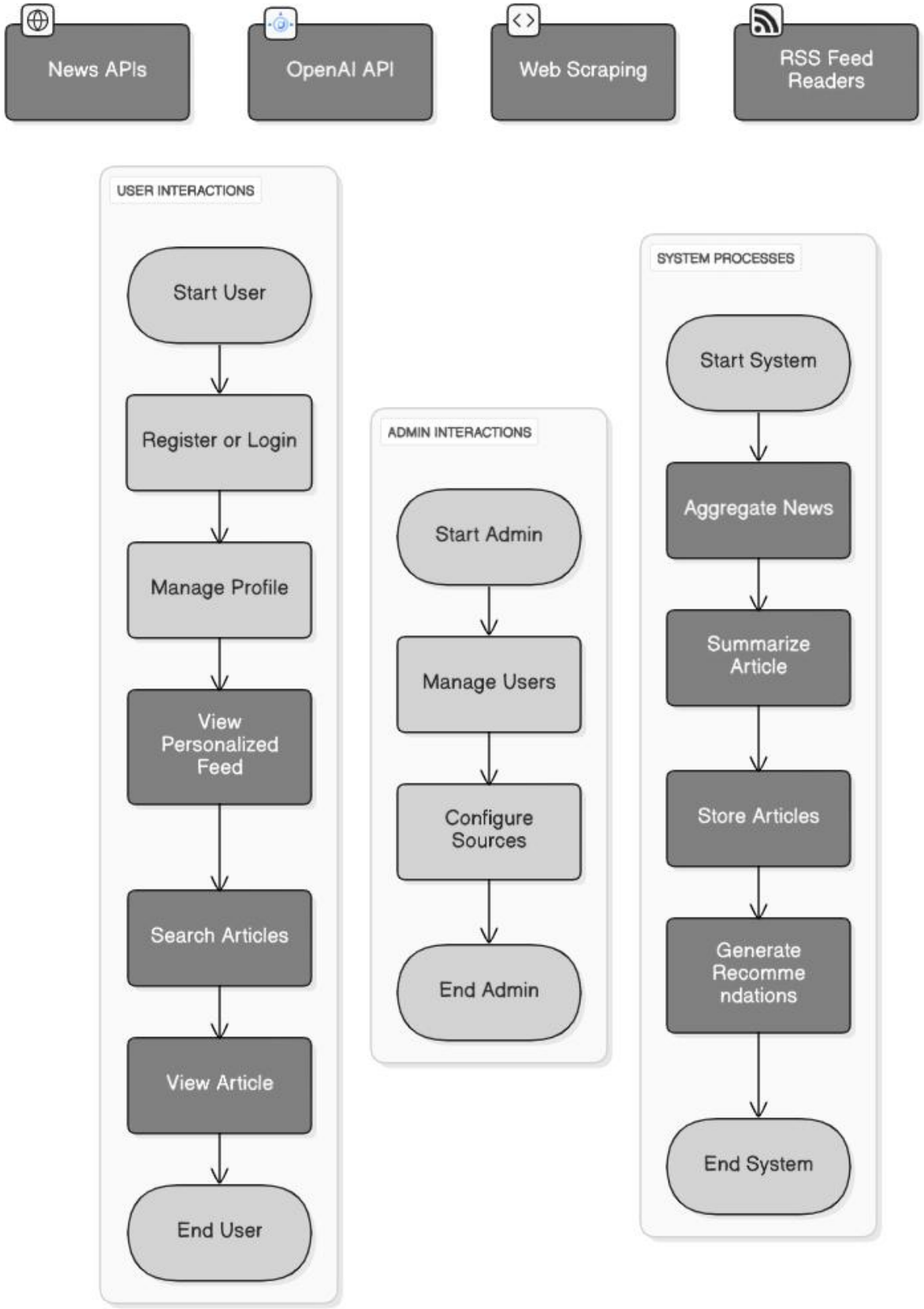

Fig 8.3 Use Case Diagram

- **UML Diagram**

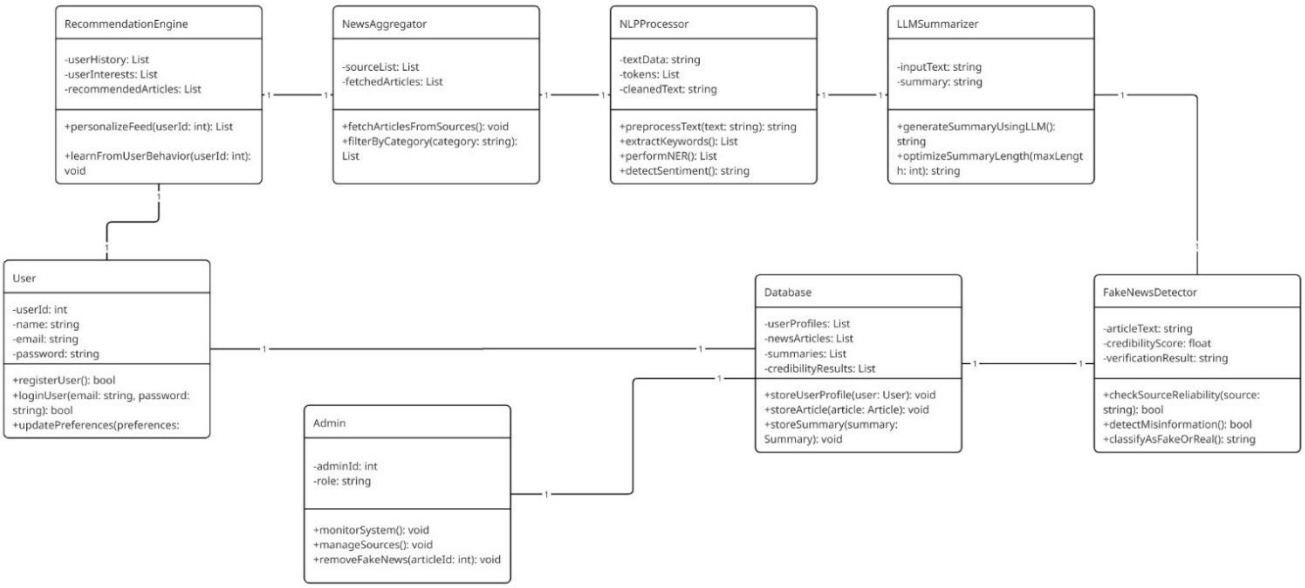

Fig 8.4 UML Diagram

9. System implementation-code documentation

SYSTEM IMPLEMENTATION - CODE DOCUMENTATION

1. Frontend Implementation

The frontend is developed using **React.js 18** and **Tailwind CSS**, offering a responsive and intuitive interface. The dashboard allows users to:

- View personalized news feed with recommended articles
- Access category-wise news (Politics, Technology, Sports, Entertainment, Health, Business)
- Search and filter articles by date, source, category, and credibility score
- Bookmark articles and track reading history
- Rate articles and manage preferences (categories, sources, summary length)
- Visualize reading analytics through charts and statistics

The interface communicates with the backend through **RESTful APIs** using Axios, implementing JWT-based authentication for secure data exchange.

2. Backend Implementation

The backend is developed using **Flask (Python)**, which handles all business logic and integrates with:

- **Web scraping module** (BeautifulSoup4, Scrapy) for news collection from 15+ sources
- **ML module** (SVM, Random Forest) for fake news detection with 85%+ accuracy
- **LLM integration** (BERT/GPT-2 via Hugging Face Transformers) for abstractive summarization
- **Personalization engine** using hybrid filtering (collaborative + content-based)
- **External APIs** (OpenAI, News API, MediaStack) for enhanced functionality

The backend uses **APScheduler** that checks for new articles every 30 minutes, runs fake news detection, generates summaries, and updates recommendation models daily.
Key API Endpoints: /api/auth/login, /api/news/feed, /api/news/search, /api/news/summarize, /api/user/preferences

3. Database Implementation

The system uses a **hybrid database architecture**:
MongoDB (NoSQL): Stores unstructured article data

- articles - Complete article content, metadata, URLs
- summaries - Generated summaries (brief, medium, detailed)
- sources - News source information and credibility scores

PostgreSQL (SQL): Stores structured user and relational data

- users - User credentials and profile information
- user_preferences - Category and source preferences
- reading_history - Article reading logs with timestamps

- bookmarks - Saved articles
- ratings - User ratings and feedback

Redis: Caching and session management for JWT tokens and frequently accessed data
The database ensures data security using **AES-256 encryption** for sensitive information and implements automated daily backups.

4. AI and Machine Learning Integration

Fake News Detection: An ensemble ML approach evaluates articles using TF-IDF features, word embeddings, and linguistic patterns to classify authenticity:

- **High credibility (80-100)** → Reliable, authentic
- **Medium credibility (50-79)** → Neutral, requires verification
- **Low credibility (0-49)** → Suspicious, potential fake news

LLM Summarization: Fine-tuned BERT/GPT-2 models generate abstractive summaries in three lengths (brief, medium, detailed), reducing article length by 70-80% while maintaining 90% information retention.

Personalization: Hybrid recommendation system combining content-based filtering (TF-IDF, cosine similarity) and collaborative filtering (user-user similarity) with 60-40 weighting. The system introduces 20% diverse content to prevent echo chambers.

5. Testing and Deployment

Testing: Unit testing (pytest), integration testing (Postman), performance testing (Locust for 1000+ concurrent users), and security testing (SQL injection, XSS prevention) were conducted. User acceptance testing achieved 80%+ satisfaction.

Deployment: The application is hosted on **AWS/GCP/Azure** using Docker containerization, Nginx web server, and Gunicorn WSGI server. CI/CD pipeline implemented via GitHub Actions ensures automated testing and deployment. Monitoring tools (Sentry, ELK Stack) track system health and performance.

10. Project Plan.

Phase 1: Research and Analysis (August - September 2025)

Week 1-2:

- Literature review and technology research
- Market analysis and competitor study
- Requirement gathering and documentation

Week 3-4:

- System design and architecture planning
- Technology stack selection
- Database schema design

Week 5-6:

- Development environment setup
- Initial prototype development
- Proof of concept implementation

Week 7-8:

- Architecture review and refinement
- Resource allocation and timeline adjustment

Phase 2: Core Development (October - December 2025)

October:

- Data collection module development
- Web scraping implementation
- Database setup and integration

November:

- Fake news detection model training
- Text preprocessing pipeline development
- Initial ML model implementation

December:

- LLM integration for summarization
- User management system
- Basic frontend development

Phase 3: Advanced Features (January - February 2026)

January:

- Personalization engine development
- Recommendation system implementation
- Advanced UI/UX design

February:

- System integration and testing
- Performance optimization
- Security implementation

Phase 4: Testing and Deployment (March 2026)

Week 1-2:

- Unit testing and integration testing
- User acceptance testing
- Performance and security testing

Week 3-4:

- Bug fixes and optimization
- Documentation completion
- Final presentation preparation

11.Conclusion

The Personalized News Aggregator offers a modern solution to the challenges of information overload and misinformation in digital news consumption.

- By leveraging Large Language Models for intelligent summarization and Machine Learning for fake news detection, this system moves beyond generic news feeds.
- The personalization engine ensures that content delivery is relevant yet diverse, helping to maintain user engagement while preventing echo chambers.
- Ultimately, this project aims to create a smart, reliable, and user-friendly tool that automates news curation, improves information quality, and enhances the news consumption experience.

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